Longo, S., Clausen, R. & Clark, B. *The Tragedy of the Commodity: Oceans, Fisheries and Aquaculture*. New Brunswick: Rutgers University Press, 2015.

Biologist Garret Hardin (1968), in his most-discussed essay, *The Tragedy of the Commons,* presented an intuitive argument about supposedly flawed human nature to explain the ruin of the ecological commons. The flaw in human nature, Hardin believed, is human egoism, which is responsible for depleting the commons, (*i.e*., community grazing grounds, grasslands, forests and public lands). For decades, environmental sociologists have been critically engaging with Hardin's subjective logic of human behavior and have garnered evidence that contradicts the depletion of which Hardin spoke—in the form of thriving natural reserves on common property—which, they assert, is attributable to a very different sort of human nature than the one Hardin described: one that is inherently cooperative, communal, collectivist, and altruistic. Yet the intuitive appeal of Hardin's argument remains undented, especially in capitalist economies. It is tempting to say that Hardin's essay has attained the same status and retained the same unflagging influence as Thomas Malthus's (1798) *An Essay on Population*. While Hardin saw depletion of resources as arising from human egoism, Malthus identified it as stemming from the 'out-of-control carnal pleasures' of the impoverished and the exponential population growth resulting therefrom. Malthus contended that the ever-soaring growth in human population was unsustainable because of environmental constraints on food production. In his reckoning, population grows geometrically, while food grows arithmetically. As both cannot keep pace with each other, growing human numbers will ultimately outrun food supplies, leading to famines, epidemics, wars, and eventual population crash. Malthus met giants among his critics such as Frederick Engels, who vigorously debunked Malthus's "population theory" as "vile and infamous doctrine," and "a repulsive blasphemy against man and nature" (Bell, 2011, p.96) that is retrospectively found to be racist (Wichterich, 1988).

Hardin Gets His Share of Critics Too

Hardin escaped Malthus's fate until a trinity of sociologists came together to definitively challenge Hardin's assumption, which still begs empirical validation. Stefano Longo, Rebecca Clausen, and Brett Clark, in *The Tragedy of the Commodity: Oceans, Fisheries and Aquaculture*, offer an evidence-based rebuttal to Hardin's (at best) oversimplified argument. Their rebuttal sets them apart from two generations of sociologists who challenged generalities within Hardin's facile assumption, though mostly in passing.[1] Authors of the *Tragedy of the Commodity* move far beyond just punching holes in Hardin's assumption about human nature and successfully offer an alternative theory of why and how the world's commons are being depleted.

'Tragedy of the Commons' Thesis is Born

The authors argue that it is not the communal ownership of the commons that is at the root of ecological degradation and depletion, as was assumed by Hardin. Rather, it is the commodification of the commons that is wreaking havoc on natural reserves. The authors trace the intellectual genesis of Hardin's thesis to fishery economists' analyses of the 1950s. Of these, they cite the works of H. Scott Gordon, Anthony Scott, and Milner Schaffer as having shaped Hardin's thinking on common property resources. This body of works collectively advocated property rights in the commons, which finally came to found Hardin's tragedy of the commons thesis. In 1954, Gordon applied economic analysis to fisheries as a common property resource and concluded that the natural resources of the sea yield no economic rent because they are not

'private property.' As such, he asserted, "The rent such resources may yield is not capable of being appropriated by anyone" (Longo et al., 2017, p. 28). A year later, Anthony Scott further boosted the economic logic of private property rights in the commons as an instrument of the commons' management. In the same vein, Schaffer laid the groundwork for the concepts of 'maximum sustainable yield,' and 'maximum economic yield' in fisheries and concluded that "free access to fishery by all citizens and the obtaining of the possible economic yield are mutually exclusive" (Longo et al., p. 28).

Hardin built on these analyses, distilling their insights and formulating his tragedy of the commons thesis. Hardin "suggested that land, as well as other natural resources that are common property, will be degraded by the competing individual interests of the users. Only control or coercion by private entities or the state can stem the inevitable destruction of commonly held nature" (Longo et al., 2017, p. 28). But the analyses upon which Hardin built his own thesis failed to see ecological ruin as a result of commodification, which is the primary feature of private property. Ironically, the individualism and egoism that private property generates in commodification goes completely unnoticed both in classical and contemporary economic analysis. Commodification, as Longo, Clausen, and Clark argue, does not necessarily serve social needs. It is rather fueled by the bottomless appetite for profits, ultimately commodifying the entire planet, piece by piece, parcel by parcel. The result is one depleted ecosystem after another, which is a social structural feature of the system that governs the economy and society: capitalism. This conclusion is in stark contrast to Hardin's and his explanation of the tragedy of the commons as being attributed to the individualistic self-interest of commons users, as embodied in private property rights.

The Commons or Commodification? Which Explains Ecological Depletion?

These contrary perspectives entail two diametrically opposed implications of vital concern to the commons' management. Hardin's argument for the privatization of common property resources, in order to conserve them, constitutes the pivot of capitalism. In contrast, Longo, Clausen, and Clark observe in privatization the very ruin of commons. They argue for a retreat from the commodification of commons (*i.e*., leaving more natural reserves as public property) as a way forward to their conservation. Delving deep into this line of argument, they demonstrate that it is the current set of human relationships with nature in general, and the commons in particular, that is wreaking havoc on their degradation and depletion. Human overproduction, overconsumption, commodification, and marketization of natural commons must be recast to bring them into balance with ecological limits. The authors argue that this re-balance cannot be achieved without reconsidering the ways and means by which humans relate to nature and the commons.

Social Metabolism and Ecological Rift

Longo, Clausen, and Clark appropriate Karl Marx's concept of social and ecological metabolism—referring to the flow of energy and materials between nature and society—a concept which was reintroduced in its contemporary iteration by one of the leading ecological thinkers of our time, John Bellamy Foster (1999; Foster et al., 2011), who, among others, is credited with discovering a 'Marx the Ecologist,' and thus making a memorable contribution to eco-Marxism. Longo, Clausen, and Clark use this concept for a critical review of contemporary nature-society relations under capitalist commodification. They argue these relations are unstable due to the current state of social metabolism: Society is depleting nature by taking more of its

resources than it can replenish, and degrading it by returning more of waste than it can absorb. This ecological rift is thus the result of social metabolic relations under capitalist commodification.

While Hardin deploys speculative judgments, Longo, Clausen, and Clark build a systematic and objective analysis to challenge Hardin's 'commonsense' and intuitive argumentation. Their broader empirical case involves that of the marine ecosystem and its social metabolism over, in the particular case of bluefin tuna, a thousand years. They date modern industrialized fishing to the nineteenth century and its subsequent expansion to the post-war years of the twentieth century. Capital investment and technological advances have since multiplied the intensity of fishing operations. As a result, "Global captures increased more than fourfold between 1950 and 2000, from 20 million tons to about 90 million tons" (p. 4). The authors attribute rising catch to commercial fishing, which uses three major operational technologies: trawlers, longline, and purse seines. Of these, they blame bottom trawling for by-catch, *i.e.*, "unintentionally harvested fish that are unwanted or lack a market." In their estimation, one-third of all catch in the United States turns up as by-catch, which gets tossed away as waste.

Bluefin Tuna and Pacific Salmon Get Commodified

Longo, Clausen, and Clark are not necessarily opposed to modern methods of fishing per se, but they situate these methods within their social and historical context to show how they have contributed to the commodification and commercialization of fish production, which, in turn has played a part in destabilizing marine ecology. The authors deploy their 'tragedy of the commodity' explanatory framework to provide a point of departure from Hardin's theory and create space for innovative ways of understanding ecological transformations, especially ones that are disruptive of marine ecosystems. The authors bring insights from their decade-long research on oceans, fisheries, and aquaculture to bear on their path-breaking theoretical approach, which is a paradigmatic contribution to a new area of study, which they name 'marine sociology,' as well as environmental sociology.

Their case studies—on bluefin tuna and Pacific salmon, which make up chapters 4 and 5 respectively—are grounded in socio-historical perspectives, with which the authors examine a fundamental shift in the relationship between humans and marine and aquaculture systems. They conduct a thousand-year review of the bluefin tuna fishery in the Mediterranean to comprehend the socio-ecological processes of production there, which they found coevolved and culminated in a coastal trapping system that was central to Mediterranean economies and cultures. They then document contemporary shifts in culture, politics, economics, and technology that have left an expanding "capital footprint" on fishing, particularly since the 1970s. These transformations co-occurred with the burst of global trade in sushi that made bluefin tuna the world's most prized fish. The global commodification of bluefin tuna, thus, contributed to the collapse of traditional fisheries, reorganization of labor, and harvesting of fish. The authors connect these findings with the growth of "tuna ranches" in the Mediterranean, a system of fishing that became highly controversial due to its social and ecological cost.

In the companion case of fishing Pacific salmon, Longo, Clausen, and Clark examine how the commodification process shaped harvesting practices in Pacific salmon fishing and led to the decline of salmon fisheries. They then critique Hardin's tragedy of the commons thesis, which was originally deployed as an explanation meant to facilitate salmon restoration but which resulted in a myriad of unintended outcomes. Furthering this analysis, the authors explore how

salmon farming was billed as an answer to declining fish populations, a means for economic development, and a way to produce food in a more ecologically friendly manner. Ultimately, however, the farming of the treasured Pacific salmon led to none of these outcomes.

Both case studies further our social scientific understanding of the depletion of global fish stocks and highlight the socio-ecological contradictions that afflict modern aquaculture. The empirical evidence marshaled within these analyses powerfully substantiates the ecological rift that social metabolism has produced worldwide on the seas and in the oceans.

Expanding 'Terra-ist' Concern to Marine Ecology

With their sound empirical analysis, Longo, Clausen, and Clark make an unprecedented contribution to the evolution of environmental sociology, which, for decades, has limited itself to the study of terrestrial ecosystems. With a shift in focus to the marine system, Longo, Clausen, and Clark discover a blind spot in environmental sociology the size of three-fourths of the planet. (With three quarters of the Earth's surface being ocean, efforts to maintain the planet's ecosystems and environment should perhaps be called "blue," not green.) They argue the earth's terrestrial and aquatic systems are integrated in the ecological web of life. Fixing or keeping the terrestrial system healthy cannot help restore planetary balance unless the ailing aquatic system is simultaneously tended to. The marine ecosystem is humanity's last resort. Yet the Earth's oceans are being commodified and metabolized faster than terra firma.

Longo, Clausen, and Clark thus expand the 'terra-ist' concern to the marine environment to square the circle. They call this expansion 'marine sociology.' This widening of 'terra-ist' concern represents a paradigm shift, which makes their contribution paradigmatic in nature. This shows the continued opportunity for exploration within the sub-discipline of environmental sociology, which itself grew out of a foundational critique of mainstream sociology and its purported neglect of the biophysical environment since the onset of the industrial revolution. Although the field of environmental sociology has been quite productive, it remained centered on the terrestrial environment until the *Tragedy of the Commodity* alerted us to its blind-spot: neglect of the marine environment. The ultimate goal that Longo, Clausen, and Clark set for themselves was to chart a way forward to recovery and restoration of what they call the 'World Ocean.' They see a glimpse of this success in Marine Protected Areas (MPAs) on the high seas. Much more, however, remains to be done, especially in terms of the global governance of the oceans and the high seas, enforcement of the law of the sea, protection of the Arctic, defusing of a global scramble for the parceling out the seas and oceans into national jurisdictions, and most importantly, addressing the acidification and pollution of ocean waters. *The Tragedy of the Commodity*, in this context, is a clarion call for recalibrating the world's social metabolism and healing the ecological rift that has degraded and depleted its seas and oceans, and, finally, for turning the commodified economy of capitalism into the 'blue' economy of the planet.

[1] Among social scientists, one notable exception to this is the Nobel Laureate economist Elinor Ostrom (1990), who was a friend of Hardin but disagreed with his one-dimensional assumptions about human nature and used a systematic and scientifically-based approach to debunk his argument.

Tarique Niazi PhD
University of Wisconsin-Eau Claire
NIAZIT@uwec.edu

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